

Poisson Regression, Three Ways

Maximum Likelihood, Penalization, and Bayesian MCMC on a Count Model

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Abstract

We fit a single, deliberately simple model — a Poisson generalized linear model for library thesis-checkout counts — under three estimation philosophies and compare what each one says. The frequentist pass derives the likelihood, maximizes it numerically, and builds confidence intervals from asymptotic theory and from the bootstrap. The penalized pass adds a ridge penalty to the slope and traces the resulting shrinkage path. The Bayesian pass places moderately informative Gaussian priors on the coefficients and samples the joint posterior with a hand-coded Metropolis algorithm, complete with convergence diagnostics. Under weak priors and $n = 85$ observations, all three estimators nearly agree — and seeing precisely *how* they agree, and where they would not, is the point of the exercise.

1 Introduction

Every semester in Math 141, Reed students collect a sample of data from the library’s thesis tower, recording for each senior thesis its *age* (in years) and the number of times it has been *checked out*. The response is a count, so ordinary least squares is the wrong tool: a straight line happily predicts negative checkouts, and the constant-variance assumption fails for counts, whose spread grows with their mean. The natural model is **Poisson regression**, a generalized linear model (GLM) for count data.

This report fits that one model three ways — by maximum likelihood, by penalized maximum likelihood, and by Bayesian inference via Markov chain Monte Carlo — and compares the estimates, the intervals, and the reasoning behind them. It is a revised, standalone treatment of a problem originally posed on the MATH 392 final exam; the analysis has been re-derived, re-run, and corrected end to end.

2 The Data

The sample contains $n = 85$ theses. Ages run from 1 to 59 years (mean 22.6); checkouts run from 0 to 10, with 33 theses — 39% — never checked out at all. Figure 1 shows the (jittered) data: a cloud hugging zero, with a hint that older theses have accumulated more checkouts.

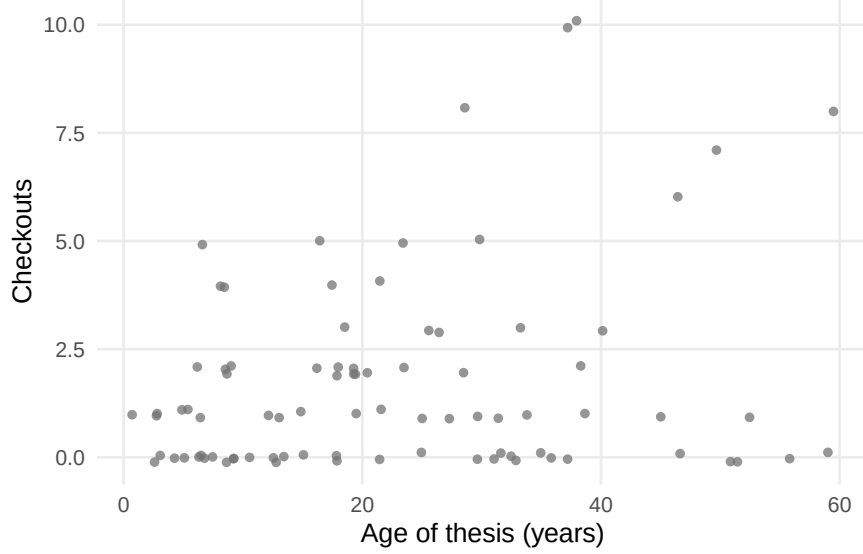


Figure 1: Checkouts against thesis age. Points are jittered to relieve overplotting of the many tied counts.

3 The Model

Poisson regression models a count response Y given a predictor matrix X (here $n \times 2$: an intercept column and age) as

$$Y_i | x_i \sim \text{Poisson}(\lambda_i), \quad \lambda_i = e^{x_i^\top \beta} = e^{\beta_0 + \beta_1 \text{age}_i},$$

with the observations conditionally independent. Like every GLM, it has three components:

1. **A response distribution.** $Y_i | x_i \sim \text{Poisson}(\lambda_i)$, so $P(Y_i = y) = \lambda_i^y e^{-\lambda_i} / y!$ on the non-negative integers.
2. **A linear predictor.** $\eta_i = x_i^\top \beta = \beta_0 + \beta_1 \text{age}_i$.
3. **A link function g** connecting the conditional mean to the linear predictor: $g(E(Y_i | x_i)) = \eta_i$. For Poisson regression the canonical link is the **logarithm**, $g(\lambda) = \log \lambda$; its inverse $g^{-1}(\eta) = e^\eta$ maps the linear predictor back to the mean and guarantees $\lambda_i > 0$ for *any* real coefficients.

That last clause matters: because the exponential inverse link already enforces positivity of the rate, the coefficients β_0, β_1 themselves live on the whole real line. We will lean on this when choosing priors in Section 6.

4 Maximum Likelihood

4.1 The likelihood and its maximizer

Writing $\lambda_i = e^{x_i^\top \beta}$ — each observation has its *own* rate, determined by its age — the log-likelihood of the sample is

$$\begin{aligned} \ell(\beta) &= \log \prod_{i=1}^n \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!} = \sum_{i=1}^n (y_i \log \lambda_i - \lambda_i - \log y_i!) \\ &= \sum_{i=1}^n \left(y_i x_i^\top \beta - e^{x_i^\top \beta} \right) - \sum_{i=1}^n \log y_i!. \end{aligned}$$

Differentiating gives the score equations

$$\nabla \ell(\beta) = \sum_{i=1}^n (y_i - e^{x_i^\top \beta}) x_i = X^\top (y - e^{X\beta}) = 0,$$

which are transcendental in β — no closed form exists, and we must optimize numerically. The Hessian, $-X^\top \text{diag}(\lambda_i) X$, is negative definite whenever X has full column rank, so ℓ is strictly concave: *if* a maximizer exists it is unique, and any hill-climbing routine will find it.

4.2 Numerical optimization

We code the log-likelihood directly and hand it to a Newton-type optimizer (`maxLik`), then check the answer against R's built-in `glm()`:

```
X <- cbind(1, theses$age)
Y <- theses$checkouts

log_lik <- function(B, X, Y) {
  eta <- X %*% B                                # linear predictor, one per thesis
  sum(Y * eta - exp(eta) - lfactorial(Y))
}

B_mle <- maxLik(log_lik, start = c(0, 0), X = X, Y = Y)$estimate
fit <- glm(checkouts ~ age, data = theses, family = "poisson")
```

Table 1: The two routes to the MLE agree to six decimals.

	$\hat{\beta}_0$	$\hat{\beta}_1$
hand-coded likelihood + <code>maxLik</code>	0.07121	0.019145
<code>glm(family = poisson)</code>	0.07121	0.019145

The fitted regression function is $\hat{E}(Y \mid \text{age}) = \exp(0.0712 + 0.0191 \cdot \text{age})$, drawn over the data in Figure 2. On the response scale the slope says checkouts grow by a factor of $e^{\hat{\beta}_1} = 1.019$ per year of age — about 21% per decade. Real, but gentle.

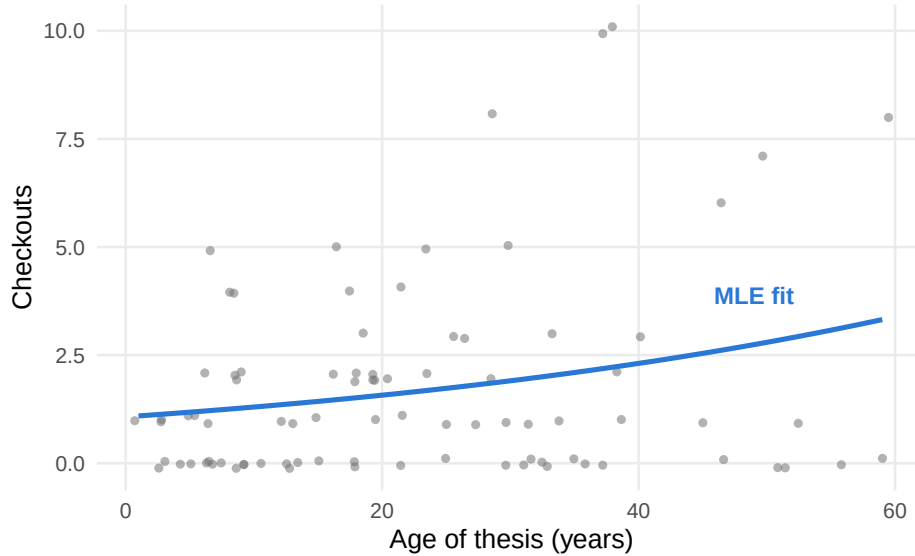


Figure 2: The maximum-likelihood fit. The curve is the estimated mean function, $\exp(\hat{\beta}_0 + \hat{\beta}_1 x)$.

4.3 A confidence interval from asymptotic theory

Under standard regularity conditions the MLE is consistent and asymptotically normal, $\hat{\beta} \sim \mathcal{N}(\beta, I(\hat{\beta})^{-1})$, where $I(\beta) = X^\top \text{diag}(\lambda_i) X$ is the Fisher information. The standard error printed by `summary(fit)` is the square root of the corresponding diagonal entry of $I(\hat{\beta})^{-1}$, so $(\hat{\beta}_1 - \beta_1)/\widehat{\text{SE}}$ is asymptotically standard normal. Inverting that pivot gives the **Wald interval** $\hat{\beta}_1 \pm z_{0.975} \widehat{\text{SE}}$, which covers the true β_1 with probability approaching 95% as $n \rightarrow \infty$. For our data:

$$\beta_1 \in [0.0091, 0.0292].$$

(R's `confint()` returns the slightly different *profile-likelihood* interval, $[0.009, 0.0291]$ — at this sample size the two are nearly indistinguishable.)

4.4 A confidence interval from the bootstrap

Asymptotic normality is a large-sample promise; with $n = 85$ and a zero-heavy response it is worth checking against a method that makes neither a normality nor a model assumption about the sampling distribution. We use the **nonparametric pairs bootstrap**: resample the (x_i, y_i) pairs with replacement, refit the GLM, and repeat.

```
set.seed(392)
n_boot <- 5000
boot_b1 <- replicate(n_boot, {
  idx <- sample(n, replace = TRUE)
  coef(glm(checkouts ~ age, data = theses[idx, ], family = "poisson"))[2]
})
boot_ci <- quantile(boot_b1, c(0.025, 0.975))
```

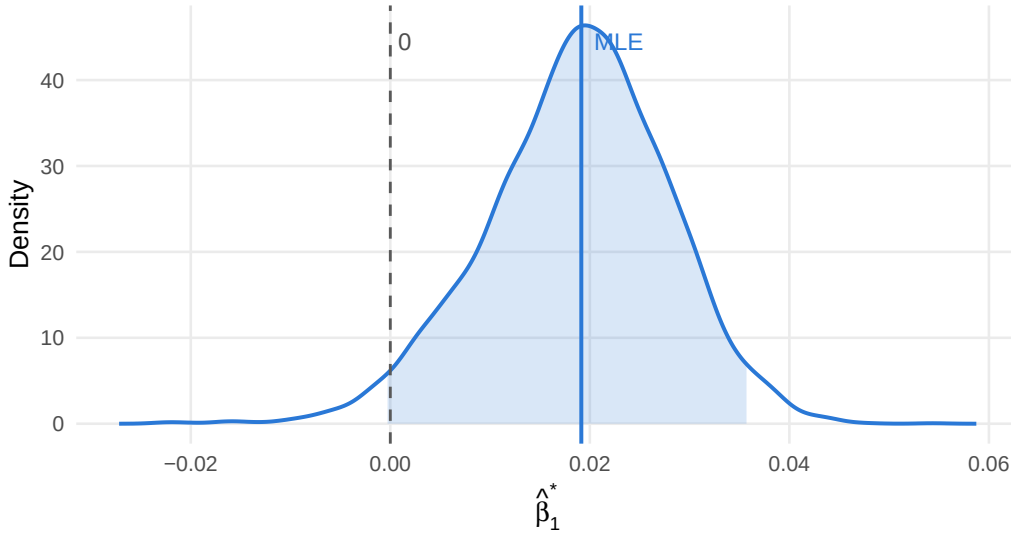


Figure 3: Bootstrap sampling distribution of $\hat{\beta}_1$ (5,000 resamples). The shaded region is the central 95%; the blue line is the MLE from the full sample.

The bootstrap percentile interval is $\beta_1 \in [-3 \times 10^{-4}, 0.0358]$ — noticeably wider than the Wald interval, with a left tail that just grazes zero (2.7% of resamples produced $\hat{\beta}_1^* \leq 0$). The asymmetry has a concrete source: a third of the theses have zero checkouts at every age, so resamples that happen to load up on old-but-never-borrowed theses can flatten the fitted slope. Still, both intervals put essentially all their mass above zero. Are we convinced the parameter is positive? At the 95% level, yes — with the honest caveat that the evidence is comfortable rather than overwhelming.

5 Penalized Likelihood

5.1 Why penalize?

Regularization (or *shrinkage*) trades a little bias for a reduction in variance by subtracting a penalty from the log-likelihood before maximizing. The classic use cases are many or collinear predictors, separation-style instability, or any setting where the unpenalized MLE has high variance. With one predictor and $n = 85$ observations we are not really in that scenario — the point here is to *watch the machinery work*: to see exactly how the estimate moves as the penalty grows.

We use a **ridge** penalty on the slope only,

$$\ell_\lambda(\beta) = \ell(\beta) - \lambda\beta_1^2,$$

leaving the intercept unpenalized, as is standard: β_0 absorbs the overall scale of the response, and shrinking it toward zero would distort every fitted value rather than regularize a slope effect. Note the penalty is subtracted **once**, outside the sum over observations — a detail that is easy to fumble in code, where placing it inside the likelihood loop silently multiplies the intended λ by n .

```
log_lik_pen <- function(B, X, Y, penwt) log_lik(B, X, Y) - penwt * B[2]^2

ridge_fit <- function(penwt)
  maxLik(log_lik_pen, start = c(0, 0), X = X, Y = Y, penwt = penwt)$estimate
```

5.2 The shrinkage path

Figure 4 traces $\hat{\beta}_1(\lambda)$ as the penalty sweeps four orders of magnitude, together with the fitted curve at the illustrative value $\lambda = 5,000$, where the slope has been shrunk from 0.0191 to 0.0151. Two things are worth noticing. First, the response to the penalty is initially almost imperceptible — the likelihood is sharply curved near the MLE, so small penalties barely move a well-identified slope. Second, as the slope shrinks the intercept *rises* (from 0.071 to 0.178): the fitted curve must still pass through the heart of the data, so what the slope gives up, the intercept absorbs.

There is also a preview of Section 6 hiding here: maximizing $\ell(\beta) - \lambda\beta_1^2$ is exactly *maximum a posteriori* estimation under a Gaussian prior $\beta_1 \sim \mathcal{N}(0, \frac{1}{2\lambda})$. Ridge regression is Bayesian shrinkage wearing a frequentist coat.

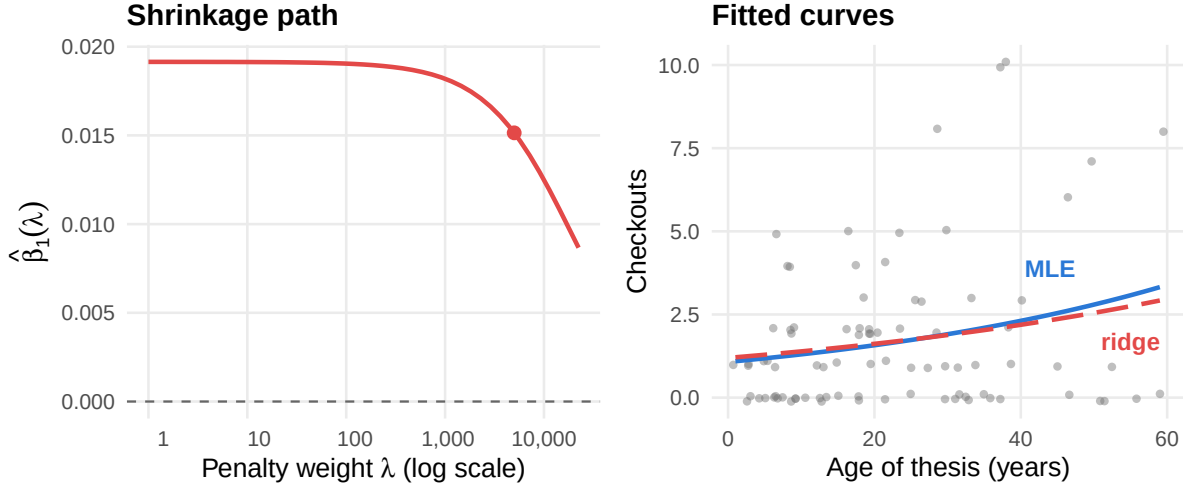


Figure 4: Left: the ridge shrinkage path for the slope; the dot marks the display value $\lambda = 5,000$. Right: the penalized fit against the MLE.

6 Bayesian Inference

6.1 Priors

Bayesian modeling asks us to say, before seeing the data, what coefficient values are plausible. The scale is awkward to intuit because the coefficients act through the exponential inverse link, so we reason on the response scale. We take independent, moderately informative Gaussians:

$$\beta_0 \sim \mathcal{N}(0, 1^2), \quad \beta_1 \sim \mathcal{N}(0, 0.05^2).$$

Sanity check. At the average age ($\bar{x} \approx 23$ years) the implied prior on the linear predictor $\eta = \beta_0 + \beta_1\bar{x}$ is centered at 0 with standard deviation $\sqrt{1 + (\bar{x} \cdot 0.05)^2} \approx 1.51$, which on the response scale spans expected checkout counts from roughly $e^{-3} \approx 0.05$ to $e^3 \approx 20$ within ± 2 standard deviations. That comfortably brackets any plausible reality for library checkouts without dogmatism — “moderately informative” in the intended sense. We model the two coefficients as *a priori* independent for simplicity, knowing the likelihood is free to (and, we will see, emphatically does) introduce posterior correlation.

Why not a positive-support prior? An earlier version of this analysis used Γ priors, reasoning that the Poisson rate must be positive. But the positivity of $\lambda = e^{x^\top \beta}$ is already guaranteed by the link for any real β — the coefficients themselves need no such constraint, and the frequentist interval for β_0 spans negative values, so forcing $\beta_0 > 0$ builds in a belief the data do not share. The choice also has a computational sting: a $\Gamma(a, b)$ prior with shape $a < 1$ has unbounded density at 0, giving the posterior an integrable singularity on the

boundary where a random-walk sampler gets stuck — re-running such a chain from different starting points gave visibly different answers for β_0 . Gaussian priors carry the intended “small effects are more plausible” information with none of those pathologies.

6.2 The posterior

By Bayes’ rule, with $\lambda_i = e^{x_i^\top \beta}$,

$$\begin{aligned} f(\beta | X, y) &= \frac{f(y | X, \beta) f(\beta)}{f(y | X)} \propto f(y | X, \beta) f(\beta_0) f(\beta_1) \\ &\propto \left[\prod_{i=1}^n \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!} \right] \exp\left(-\frac{\beta_0^2}{2 \cdot 1^2}\right) \exp\left(-\frac{\beta_1^2}{2 \cdot 0.05^2}\right). \end{aligned}$$

The normalizing constant is unavailable, which is no obstacle: the Metropolis algorithm needs the posterior only up to proportionality. We implement the three ingredients on the log scale — products of 85 densities underflow double precision, sums of logs do not:

```
s0 <- 1; s1 <- 0.05                                # prior standard deviations

log_prior <- function(theta)
  dnorm(theta[1], 0, s0, log = TRUE) + dnorm(theta[2], 0, s1, log = TRUE)

log_post <- function(theta) log_prior(theta) + log_lik(theta, X, Y)
```

6.3 The Metropolis sampler

The Metropolis algorithm performs a random walk whose long-run visit frequencies match the posterior: from the current state $\theta^{(t)}$, propose $\theta^* = \theta^{(t)} + \varepsilon$ with ε drawn from a symmetric distribution, and accept the move with probability $\min(1, f(\theta^* | y)/f(\theta^{(t)} | y))$; otherwise stay put. We use independent Gaussian increments with scales (0.12, 0.004) — roughly $0.8\times$ each coefficient’s standard error, tuned so the acceptance rate lands near the classic random-walk target of 0.3–0.45.

```
metropolis <- function(iters, prop_sd, theta0, seed) {
  set.seed(seed)
  chain <- matrix(NA_real_, iters + 1, 2, dimnames = list(NULL, c("b0", "b1")))
  chain[1, ] <- theta0
  lp_cur <- log_post(theta0)
  accepts <- 0
  for (t in 1:iters) {
    prop <- chain[t, ] + rnorm(2, 0, prop_sd)
    lp_prop <- log_post(prop)
    if (log(runif(1)) < lp_prop - lp_cur) {          # Metropolis ratio, in logs
      chain[t + 1, ] <- prop; lp_cur <- lp_prop; accepts <- accepts + 1
    } else chain[t + 1, ] <- chain[t, ]
  }
  list(chain = chain, accept_rate = accepts / iters)
}

iters <- 50000; burn_in <- 5000
run1 <- metropolis(iters, c(0.12, 0.004), theta0 = c(0, 0), seed = 92)
run2 <- metropolis(iters, c(0.12, 0.004), theta0 = c(0.5, -0.05), seed = 41)
post <- rbind(run1$chain[-(1:burn_in), ], run2$chain[-(1:burn_in), ])
```

We run **two chains** of 50,000 iterations from deliberately different starting points — $(0, 0)$ at the prior mean and $(0.5, -0.05)$ well away from it — discard the first 5,000 draws of each as burn-in, and pool the rest, leaving 90,002 posterior draws. (Discarding the burn-in is not a formality: these summaries change visibly if the transient is left in, since the far-off start drags every quantile toward its own coordinates.)

6.4 Did it converge?

Three checks, all passed:

1. **Acceptance rate.** 44.7% and 45.2% for the two chains — inside the healthy random-walk band, so the sampler is neither shuffling in place nor rejecting everything.
2. **Trace plots** (Figure 5). Both chains forget their starting points within a few hundred iterations and then oscillate stably around a common level — the “fuzzy caterpillar” of a mixing chain, with no drift and no level shifts.
3. **Agreement across chains.** Post burn-in, the two chains’ posterior means for (β_0, β_1) agree to about three decimal places $((0.071, 0.019)$ vs. $(0.073, 0.019))$, which is what independent convergence to the same distribution looks like. The autocorrelation-adjusted effective sample size is roughly 1,500 draws per chain per parameter — ample for the tail quantiles we report below.

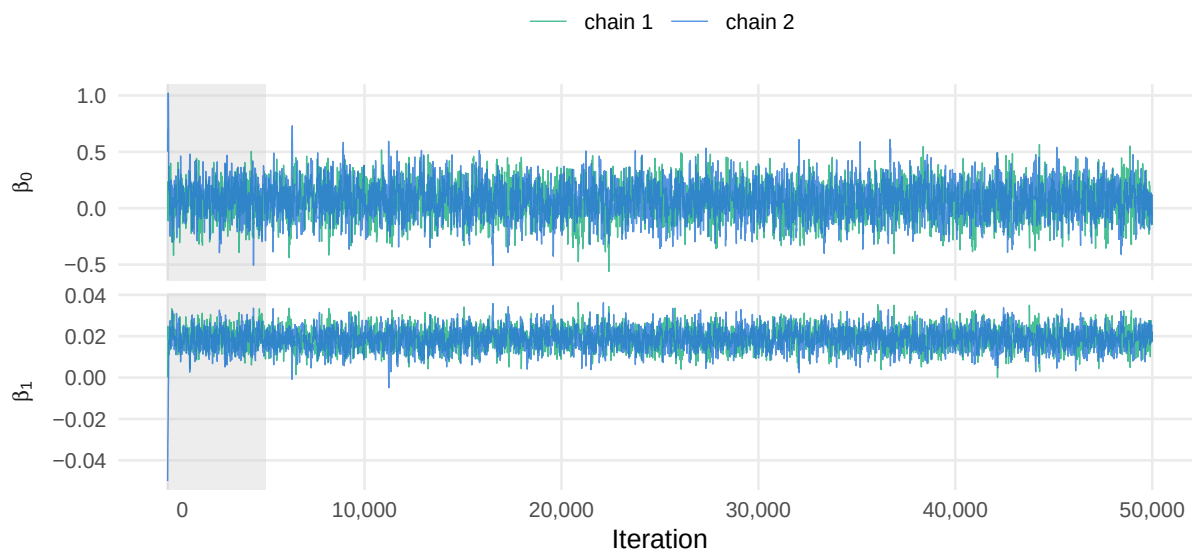


Figure 5: Trace plots for the two chains (thinned to every 10th draw for display). The shaded band is the discarded burn-in. Both chains leave their deliberately different starts and mix over the same region.

6.5 What the data did to the prior

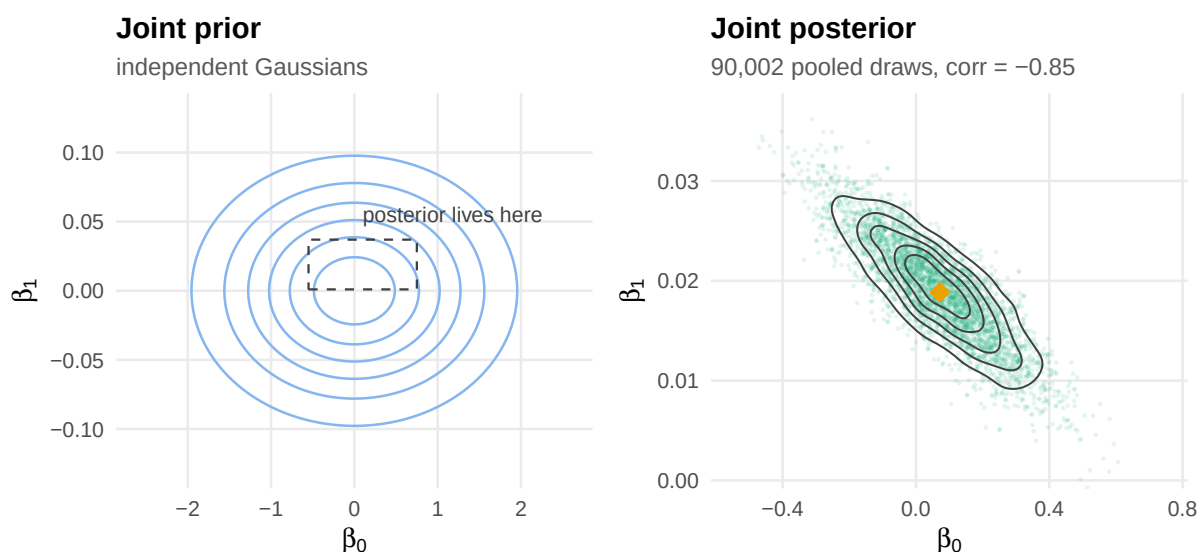


Figure 6: The joint prior (left) and joint posterior (right) over the coefficients. Note the axis scales: the small box inside the prior panel is the entire extent of the posterior panel. The diamond marks the posterior mean.

Figure 6 is the heart of the Bayesian story. The prior on the left is a broad, axis-aligned hill: the two coefficients independent, each centered at zero, jointly agnostic across a wide swath of coefficient space. The posterior on the right occupies a region roughly *one-tenth the prior's scale in each direction* — 85 observations carry a great deal of information — and it is anything but axis-aligned: the draws fall along a sharp negative ridge, with correlation -0.85 . The geometry has a tidy explanation: every thesis has positive age, so a steeper fitted slope must be paid for with a lower intercept to keep the curve passing through the data's center of mass. Independence in, correlation out — the likelihood put it there.

The marginal view (Figure 7) tells the same story one coordinate at a time: each posterior is a spike standing inside its own nearly flat prior, and each sits essentially on top of the MLE.

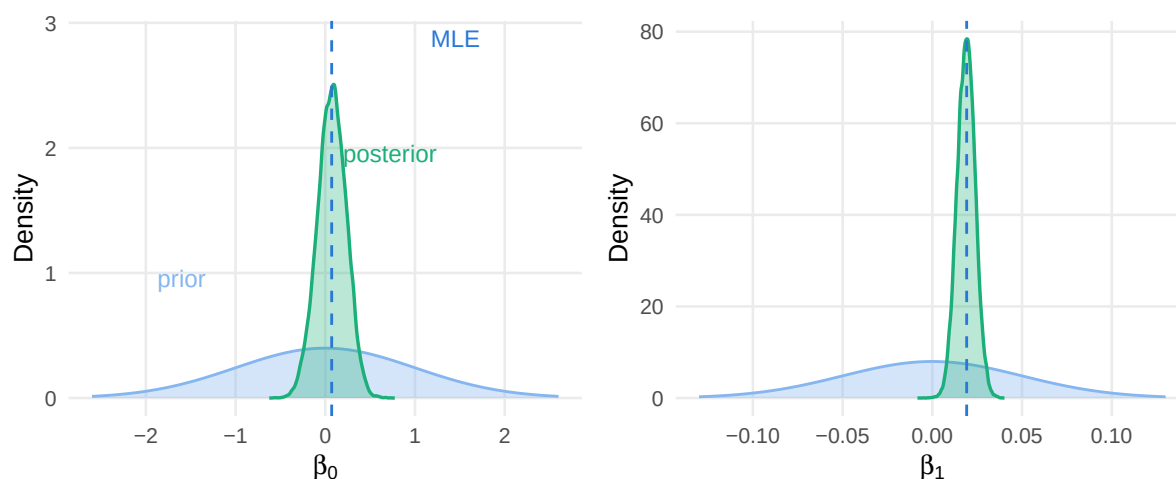


Figure 7: Marginal prior and posterior for each coefficient, with the MLE marked. The posterior concentrates in a sliver of the prior's support — the visual measure of how much the data taught us.

6.6 Credible intervals

The central 95% **credible intervals** — the 2.5% and 97.5% quantiles of the posterior draws — are

$$\beta_0 \in [-0.247, 0.376], \quad \beta_1 \in [0.009, 0.0287],$$

and the posterior probability that the slope is positive is 99.97%. Set beside the Wald interval $[0.0091, 0.0292]$, the credible interval for β_1 is a hair narrower and nudged toward zero — precisely the fingerprint of a zero-centered prior acting as gentle shrinkage — but the two essentially coincide. That is no accident: with $n = 85$ observations and deliberately weak priors, the likelihood dominates and the posterior concentrates around the MLE (the Bernstein–von Mises phenomenon). The interpretations remain philosophically distinct — the credible interval is a probability statement about β_1 given this data and prior, while the confidence interval is a coverage statement about the procedure over repeated samples — but here they land on the same numbers. Had we used sharply informative priors, or a tenth of the data, the two would visibly part ways.

7 Three Fits, One Picture

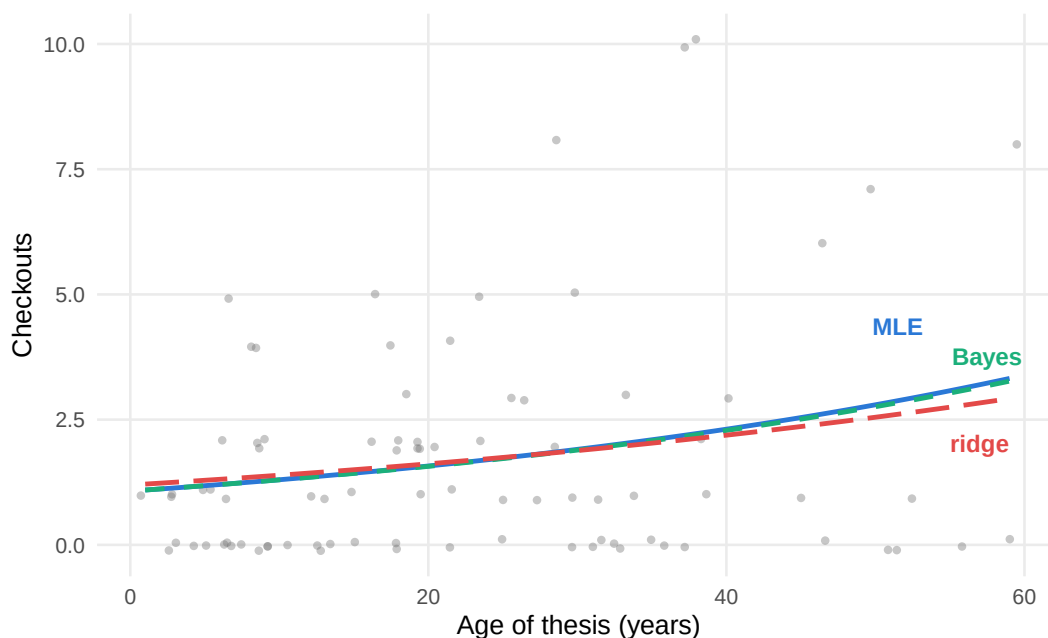


Figure 8: All three fitted mean functions. The MLE and Bayesian curves overlap almost exactly; the heavily penalized ridge fit is flatter, with a compensating higher intercept.

Table 2: The three estimates side by side.

Method	$\hat{\beta}_0$	$\hat{\beta}_1$	95% interval for β_1
Maximum likelihood	0.0712	0.0191	$[0.0091, 0.0292]$ (Wald)
Ridge ($\lambda = 5,000$)	0.1783	0.0151	—
Bayesian posterior mean	0.0722	0.0188	$[0.0090, 0.0287]$ (credible)

The three curves in Figure 8 summarize the whole report. The **MLE** and the **Bayesian posterior mean** are visually inseparable: weak priors plus a decent sample size mean the two philosophies disagree about the meaning of the estimate far more than about its value. The **ridge** fit is the deliberate outlier — with

$\lambda = 5,000$ the slope is shrunk by about 21%, and the intercept rises to compensate, pivoting the curve flatter around the data’s center of mass. None of this is mysterious: all three estimators maximize the same log-likelihood surface, differing only in what they add to it (nothing, a quadratic penalty, or a log-prior — and the middle option *is* the third option, up to a constant).

8 Conclusion

One small dataset, one model, three inferential frameworks:

- **Maximum likelihood** delivered the point estimate and, through asymptotic normality, an interval; the bootstrap corroborated it while honestly widening the left tail that the zero-heavy counts create.
- **Penalization** showed how a quadratic penalty trades bias for variance, how the intercept compensates for a shrunk slope — and that ridge is secretly a Gaussian prior.
- **Bayesian MCMC** turned broad independent priors into a tight, negatively correlated joint posterior, with a hand-coded Metropolis sampler whose convergence we could check rather than assume.

The agreement of all three answers — roughly, checkouts rise 21% per decade of thesis age — is the reassuring part. The instructive part is *why* they agree: weak priors and gentle penalties are both small perturbations of the same likelihood, and $n = 85$ is enough data for that likelihood to speak loudly.

One honest limitation deserves the last word. The Poisson model equates the conditional variance with the conditional mean, and these data push back: the Pearson statistic is 214.5 on 83 degrees of freedom, a dispersion ratio near 2.6. Checkout counts are more variable than a Poisson process would allow — some theses are simply more interesting than their age explains. A quasi-Poisson or negative-binomial model would inflate the standard errors by roughly $\sqrt{2.6} \approx 1.6\times$, and a serious application would start there. That the bootstrap interval already hinted at this — wider than the Wald interval, built as it is from the data’s actual variability rather than the model’s assumed one — is a fitting closing argument for owning more than one inferential tool.

Reproducibility

The data (`data/theses.csv`, $n = 85$) were collected by Reed College Math 141 students from the library's thesis tower and are bundled with this module. All randomness is seeded: the bootstrap uses seed 392 and the two MCMC chains use seeds 92 and 41 with starting points $(0, 0)$ and $(0.5, -0.05)$. Knitting this document re-runs every computation — the 5,000-resample bootstrap, both 50,000-iteration Metropolis chains, and all figures — from scratch.

```
sessionInfo() # R >= 4.4; packages: tidyverse, maxLik, patchwork, knitr
```